

Supporting Information

Do Large Language Models Carry Bias for Citation Recommendations?

A Case Study of Energy Research Topics

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Table S1

Outputs obtained from five different LLMs in the selected four energy related topics.

(Cells containing invalid citations are highlighted in red).

AI Engine	Topic	#	Citation
Responses Run 1 (March 17, 2026)			
Prompt: "Provide exactly 10 peer-reviewed journal citations on: new advances and challenges in (topic area)" Format each citation as: [Authors]. [Title]. [Journal], [Year], [Volume], [Pages]. DOI: [DOI]."			
ChatGPT (GPT 5.3)	Electrochemical CO ₂ Reduction	1	S. Nitopi; E. Bertheussen; S. Scott; X. Liu; A. K. Engstfeld; S. Horch; B. Seger; I. E. Stephens; K. Chan; C. Hahn; J. K. Nørskov; T. F. Jaramillo; I. Chorkendorff. Progress and Perspectives of Electrochemical CO ₂ Reduction on Copper in Aqueous Electrolyte. Chemical Reviews, 2019, 119, 7610–7672. DOI: 10.1021/acs.chemrev.8b00705.
		2	J. Han; Y. Liu; Z. Chen; X. Wang; H. Li. Advances and Challenges in the Electrochemical Reduction of CO ₂ . Chemical Science, 2024, 15, 12345–12378. DOI: 10.1039/D4SC01931H.
		3	B. Kumar. Electrochemical CO ₂ Conversion Commercialization Pathways: A Concise Review on Experimental Frontiers and Technoeconomic Analysis. Environmental Science & Technology Letters, 2024, 11, 567–576. DOI: 10.1021/acs.estlett.4c00564.
		4	F. Yu; H. Wang; Y. Zhou; X. Zhang. Electrochemical CO ₂ Reduction: From Catalysts to Reactive Interfaces. Chemical Engineering Journal, 2023, 452, 139307. DOI: 10.1016/j.cej.2022.139307.
		5	K. C. Poon; S. C. Lee; Y. Li. A Review toward Sustainable Electrochemical Reduction of CO ₂ to CO. Nanomaterials, 2022, 12, 2541–2565. DOI: 10.3390/nano12152541.
		6	K. Wiranarongkorn; S. Assabumrungrat; N. Laosiripojana. A Comprehensive Review of Electrochemical Reduction of CO ₂ to Methanol. Journal of CO ₂ Utilization, 2023, 65, 102230–102248. DOI: 10.1016/j.jcou.2023.102230.
		7	G. Leonzio. CO ₂ Electrochemical Reduction: A State-of-the-Art Review. Chemical Engineering Research and Design, 2024, 200, 123–145. DOI: 10.1016/j.cherd.2024.01.012.
		8	M. Al-Shamari; A. Al-Ajmi; S. Alenezi. Utilization of Carbon Dioxide Using Electrochemical Reduction. Results in Engineering, 2024, 22, 102345–102360. DOI: 10.1016/j.rineng.2024.102345.
		9	J. Li; S. U. Abbas; H. Wang; Z. Zhang; W. Hu. Recent Advances in Interface Engineering for Electrocatalytic CO ₂ Reduction Reaction. Nano-Micro Letters, 2021, 13, 1–35. DOI: 10.1007/s40820-021-00670-2.
		10	K. A. Alkhuzai; M. A. Alotaibi; A. Alharthi. The Role of Electrocatalysts in the Electrochemical Reduction of CO ₂ to C ₂ + Products. Catalysts, 2025, 15, 237–256. DOI: 10.3390/catal15030237.
ChatGPT (GPT 5.3)	Li-ion Batteries	1	Vega-Muratalla, V. O.; Serrano-Arévalo, T. I.; Ochoa-Barragán, R.; Ponce-Ortega, J. M. Recent Advances and Engineering Challenges of Lithium Batteries for Grid-Level Energy Storage: A Review. Industrial & Engineering Chemistry Research, 2025, 64, —. DOI: 10.1021/acs.iecr.5c03594
		2	Fazal, S.; Ahmad, F.; Khan, M. I.; Shanableh, A.; Manzoor, S. A Comprehensive Review on the Challenges Associated with Lithium-Ion Batteries and Their Possible Solutions. New Journal

			of Chemistry, 2024, 48, 14104–14127. DOI: 10.1039/D4NJ02397H
		3	Khalid, R.; Shah, A.; Javed, M.; Hussain, H. Progress and Obstacles in Electrode Materials for Lithium-Ion Batteries: A Journey towards Enhanced Energy Storage Efficiency. RSC Advances, 2025, 15, 15951–15998. DOI: 10.1039/d5ra02042e
		4	Chacana-Olivares, J.; Peceño, B.; Grageda, M.; et al. Lithium-Ion Battery Recycling: A Perspective on Key Challenges and Opportunities. npj Materials Sustainability, 2025, 3, 38. DOI: 10.1038/s44296-025-00083-7
		5	—. A Review of Recent Advances, Current Limitations, and Remedies of Lithium-Ion Batteries for Advanced Technological Applications. Future Batteries, 2025, 7, 100109. DOI: 10.1016/j.fub.2025.100109
		6	—. Recent Advances in Solid-State Batteries. Journal of the American Chemical Society, 2025, 147, 14881–14883. DOI: 10.1021/jacs.5c06058
		7	Zheng, P.; Young, D.; Yang, T.; Xiao, Y.; Li, Z. Powering Battery Sustainability: A Review of the Recent Progress and Evolving Challenges in Recycling Lithium-Ion Batteries. Frontiers in Sustainable Resource Management, 2023, 2, —. DOI: 10.3389/fsrma.2023.1127001
		8	Xu, Y.; Ge, X.; Guo, R.; Shen, W. Recent Advances in Model-Based Fault Diagnosis for Lithium-Ion Batteries: A Comprehensive Review. IEEE Transactions on Industrial Electronics, 2024, —, —. DOI: 10.48550/arXiv.2401.16682
		9	Datta, D. Electro-Chemo-Mechanical Modeling of Multiscale Active Materials for Next-Generation Energy Storage: Opportunities and Challenges. Journal of The Electrochemical Society, 2023, —, —. DOI: 10.48550/arXiv.2309.01954
		10	Gervillie-Mouravieff, C.; Bao, W.; Steingart, D. A.; Meng, Y. S. Non-Destructive Characterization Techniques for Battery Performance and Lifecycle Assessment. Nature Reviews Materials, 2023, —, —. DOI: 10.48550/arXiv.2310.15396
ChatGPT (GPT 5.3)	Perovskite Solar Cells	1	M. Noman; Z. Khan; S. T. Jan. A comprehensive review on the advancements and challenges in perovskite solar cell technology. RSC Advances, 2024, 14, 5085–5131. DOI: 10.1039/D3RA07518D.
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		3	G. Dastgeer; et al. A review on recent progress and challenges in high-efficiency perovskite solar cells. Nano Energy, 2024, 132, 104901. DOI: 10.1016/j.nanoen.2024.104901.
		4	C. Yang; et al. Achievements, challenges, and future prospects for perovskite solar cells. Light: Science & Applications, 2024, 13, 1461. DOI: 10.1038/s41377-024-01461-x.
		5	M. Afroz; et al. Perovskite solar cells: Progress, challenges, and future prospects. Solar Energy, 2025, 275, 112345. DOI: 10.1016/j.solener.2024.112345.
		6	X. Shen; et al. Key advancements and emerging trends of perovskite solar cells. Nano-Micro Letters, 2026, 18, 2022. DOI: 10.1007/s40820-025-02022-6.
		7	J. Yuan; et al. Enriching conductive capping by alkaline treatment of perovskite quantum dots towards certified 18.3%-efficient solar cells. Nature Communications, 2024, 15, 9247. DOI: 10.1038/s41467-024-53598-1.
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		9	F. Li; et al. Combining perovskites and quantum dots: Synthesis, characterization, and applications in solar cells, LEDs, and photodetectors. Advanced Energy Materials, 2022, 12, 2103564. DOI: 10.1002/aenm.202103564.
		10	H. C. Weerasinghe; et al. The first demonstration of entirely roll-to-roll fabricated perovskite solar cell modules under ambient room conditions. Nature Communications, 2024, 15, 1234–1242. DOI: 10.1038/s41467-024-XXXXX.
ChatGPT (GPT 5.3)	QD LEDs	1	Jiang, X.; Fan, Z.; Luo, L.; Wang, L. Advances and Challenges in Heavy-Metal-Free InP Quantum Dot Light-Emitting Diodes. Micromachines, 2022, 13, 709. DOI: 10.3390/mi13050709.
		2	Kim, J. Recent Progresses and Challenges in Colloidal Quantum Dot Light-Emitting Diodes: A Focus on Electron Transport Layers with Metal Oxide Nanoparticles and Organic Semiconductors. Nanoscale Horizons, 2024, 9, 2167–2197. DOI: 10.1039/D4NH00370E.
		3	Kim, J.; Roh, J.; Park, M.; Lee, C. Recent Advances and Challenges of Colloidal Quantum Dot Light-Emitting Diodes for Display Applications. Advanced Materials, 2024, 36, e2212220. DOI: 10.1002/adma.202212220.
		4	Tan, S.; Horowitz, J. R.; Tye, O. J.; Bawendi, M. G. Challenges of II–VI and III–V Blue Quantum Dot Light-Emitting Diodes. Advanced Materials, 2026, 38, e12379. DOI: 10.1002/adma.202512379.
		5	Li, B.; Chen, F.; Xu, H.; Song, Y.; Yan, X.; Xu, Q.; Wu, L.; Yan, Y.; Hou, W.; Cao, W.; Shen, H.; Fan, F. Advances in Understanding Quantum Dot Light-Emitting Diodes. Nature Reviews Electrical Engineering, 2024, 1, 412–425. DOI: 10.1038/s44287-024-00059-0.
		6	Tian, D.; Ma, H.; Huang, G.; Gao, M.; Cai, F.; Fang, Y.; Li, C.; Jiang, X.; Wang, A.; Wang, S.; Du, Z. A Review on Quantum Dot Light-Emitting Diodes: From Materials to Applications. Advanced Optical Materials, 2022, 10, 2201965. DOI: 10.1002/adom.202201965.
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		10	Colvin, V. L.; Schlamp, M. C.; Alivisatos, A. P. Light-Emitting Diodes Made from Cadmium Selenide Nanocrystals and a Semiconducting Polymer. Nature, 1994, 370, 354–357. DOI: 10.1038/370354a0.
Claude (Sonnet 4.6)	Electrochemical CO ₂ Reduction	1	Han, J.; Bai, X.; Xu, X.; Bai, X.; Husile, A.; Zhang, S.; Qi, L., & Guan, J. Advances and Challenges in the Electrochemical Reduction of Carbon Dioxide. Chemical Science, 2024, 15, 7870–7907. DOI: 10.1039/D4SC01931H
		2	Birdja, Y.Y., Pérez-Gallent, E., Figueiredo, M.C., Göttle, A.J., Calle-Vallejo, F., & Koper, M.T.M. Advances and Challenges in Understanding the Electrocatalytic Conversion of Carbon

			Dioxide to Fuels. <i>Nature Energy</i> , 2019, 4, 732–745. DOI: 10.1038/s41560-019-0450-y
		3	Wakerley, D., Lamaison, S., Wicks, J., Clemens, A., Feaster, J., Corral, D., Jaffer, S.A., Sarkar, A., Fontecave, M., Duoss, E.B., Baker, S., Sargent, E.H., Jaramillo, T.F., & Hahn, C. Gas Diffusion Electrodes, Reactor Designs and Key Metrics of Low-Temperature CO ₂ Electrolysers. <i>Nature Energy</i> , 2022, 7, 130–143. DOI: 10.1038/s41560-021-00973-9
		4	Jin, S., Hao, Z., Zhang, K., Yan, Z., & Chen, J. Advances and Challenges for the Electrochemical Reduction of CO ₂ to CO: From Fundamentals to Industrialization. <i>Angewandte Chemie International Edition</i> , 2021, 60, 20627–20648. DOI: 10.1002/anie.202101818
		5	Kiss, A.A., Pérez-Fortes, M., & Moulijn, J.A. Challenges and Opportunities for CO ₂ Electroreduction from a Process Systems Engineering Perspective. <i>Frontiers in Energy Research</i> , 2024, 12, 1340622. DOI: 10.3389/fenrg.2024.1340622
		6	Chen, C., Li, J., Tan, X., Zhang, Y., Li, Y., He, C., Xu, Z., Zhang, C., & Chen, C. Harnessing Single-Atom Catalysts for CO ₂ Electroreduction: A Review of Recent Advances. <i>EES Catalysis</i> , 2024, 2, 71–107. DOI: 10.1039/D3EY00150D
		7	Sikdar, N., Bhanja, P., & Bhaumik, A. Electrochemical CO ₂ Reduction Reaction: Comprehensive Strategic Approaches to Catalyst Design for Selective Liquid Products Formation. <i>Chemistry – A European Journal</i> , 2024, 30, e202402477. DOI: 10.1002/chem.202402477
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		9	Huang, J.E., Li, F., Ozden, A., Sedighian Rasouli, A., García de Arquer, F.P., Liu, S., Zhang, S., Luo, M., Wang, X., Lum, Y., Xu, Y., Bertens, K., Miao, R.K., Dinh, C.T., Sinton, D., & Sargent, E.H. CO ₂ Electrolysis to Multicarbon Products in Strong Acid. <i>Science</i> , 2021, 372, 1074–1078. DOI: 10.1126/science.abg6582
		10	Meng, Y., Huang, H., Zhang, Y., Cao, Y., Lu, H., & Li, X. Recent Advances in the Theoretical Studies on the Electrocatalytic CO ₂ Reduction Based on Single and Double Atoms. <i>Frontiers in Chemistry</i> , 2023, 11, 1172146. DOI: 10.3389/fchem.2023.1172146
Claude (Sonnet 4.6)	Li-ion Batteries	1	Frith, J.T., Lacey, M.J., and Ulissi, U. A non-academic perspective on the future of lithium-based batteries. <i>Nature Communications</i> , 2023, 14(1), 420. DOI: 10.1038/s41467-023-35933-2
		2	Ha, Y., Licht, R.B., Bogart, T.D., and Beers, K.M. Bridging the gap between academic and industry Li-ion battery research. <i>Frontiers in Batteries and Electrochemistry</i> , 2024, 2, 1287887. DOI: 10.3389/fbael.2023.1287887
		3	Toki, G.F.I., Hossain, M.K., Rehman, W.U., Manj, R.Z.A., Wang, L., and Yang, J. Recent progress and challenges in silicon-based anode materials for lithium-ion batteries. <i>Industrial Chemistry & Materials</i> , 2024, 2(2), 226–269. DOI: 10.1039/D3IM00115F
		4	Zheng, Y., Shen, Y., Guo, J., Li, J., Wang, J., Ning, D., Liu, Y., Huang, Y., Tang, Y., Deng, Y., and Yan, H. Recent advances in solid-state lithium batteries based on anode engineering. <i>Nano Research Energy</i> , 2024, 3, e9120118. DOI: 10.26599/NRE.2024.9120118
		5	Lakshmi, K.C.S. Beyond lithium-ion: emerging frontiers in next-generation battery technologies. <i>Frontiers in Batteries and</i>

			Electrochemistry, 2024, 3, 1377192. DOI: 10.3389/fbael.2024.1377192
		6	Zhang, G., Chen, X., Ma, Y., Huo, H., Zuo, P., Yin, G., Gao, Y., and Fu, C. Recent advances and practical challenges of high-energy-density flexible lithium-ion batteries. <i>Frontiers of Chemical Science and Engineering</i> , 2024, 18(8), 91. DOI: 10.1007/s11705-024-2444-y
		7	Wang, J., Mao, Y., Zhang, D., Feng, X., Lu, L., Han, X., Ouyang, M., and Sauer, D.U. Advances and challenges in thermal runaway modeling of lithium-ion batteries. <i>The Innovation</i> , 2024, 5(4), 100624. DOI: 10.1016/j.xinn.2024.100624
		8	Wan, H., Wang, Z., Zhang, W., He, X., and Wang, C. Interface design for all-solid-state lithium batteries. <i>Nature</i> , 2023, 623, 739–755. DOI: 10.1038/s41586-023-06679-w
		9	Adenusi, H., Hasa, I., and Passerini, S. Lithium batteries and the solid electrolyte interphase (SEI)—progress and outlook. <i>Advanced Energy Materials</i> , 2023, 13(9), 2203307. DOI: 10.1002/aenm.202203307
		10	Pham, M.T.M., Darst, J.J., Finegan, D.P., and Walker, W.Q. Challenges and opportunities for high-quality battery production at scale. <i>Nature Communications</i> , 2025, 16(1), 722. DOI: 10.1038/s41467-025-55861-7
Claude (Sonnet 4.6)	Perovskite Solar Cells	1	Liu, S., Li, J., Xiao, W., Chen, R., Sun, Z., Zhang, Y., Lei, X., Hu, S., Kober-Czerny, M., Wang, J., Ren, F., Zhou, Q., Raza, H., Gao, Y., Ji, Y., Li, S., Li, H., Qiu, L., Huang, W., Zhao, Y., Xu, B., Liu, Z., Snaith, H. J., Park, N.-G., & Chen, W. Buried Interface Molecular Hybrid for Inverted Perovskite Solar Cells. <i>Nature</i> , 2024, 632, 536–542. DOI: 10.1038/s41586-024-07723-3
		2	Chen, H., Liu, C., Xu, J., Maxwell, A., Zhou, W., Yang, Y., Zhou, Q., Bati, A. S. R., Wan, H., Wang, Z., Zeng, L., Wang, J., Serles, P., Liu, Y., Teale, S., Liu, Y., Saidaminov, M. I., Li, M., Rolston, N., Hoogland, S., Filleter, T., Kanatzidis, M. G., Chen, B., Ning, Z., & Sargent, E. H. Improved Charge Extraction in Inverted Perovskite Solar Cells with Dual-Site-Binding Ligands. <i>Science</i> , 2024, 384, 189–193. DOI: 10.1126/science.adm9474
		3	Liu, J., He, Y., Ding, L., Zhang, H., Li, Q., Jia, L., Yu, J., Lau, T. W., Li, M., Qin, Y., Gu, X., Zhang, F., Li, Q., Yang, Y., Zhao, S., Wu, X., Liu, J., Liu, T., Gao, Y., Wang, Y., Dong, X., Chen, H., Li, P., Zhou, T., Yang, M., Ru, X., Peng, F., Yin, S., Qu, M., Zhao, D., Zhao, Z., Li, M., Guo, P., Yan, H., Xiao, C., Xiao, P., Yin, J., Zhang, X., Li, Z., He, B., & Xu, X. Perovskite/Silicon Tandem Solar Cells with Bilayer Interface Passivation. <i>Nature</i> , 2024, 635, 596–603. DOI: 10.1038/s41586-024-07997-7
		4	Ugur, E., Said, A. A., Dally, P., Zhang, S., Petoukhoff, C. E., Rosas-Villalva, D., Zhumagali, S., Yildirim, B. K., Razzaq, A., Sarwade, S., Yazmaciyan, A., Baran, D., Laquai, F., Deger, C., Yavuz, I., Allen, T. G., Aydin, E., & De Wolf, S. Enhanced Cation Interaction in Perovskites for Efficient Tandem Solar Cells with Silicon. <i>Science</i> , 2024, 385, 533–538. DOI: 10.1126/science.adp1621
		5	Aydin, E., Allen, T. G., De Bastiani, M., Razzaq, A., Xu, L., Ugur, E., Liu, J., & De Wolf, S. Pathways toward Commercial Perovskite/Silicon Tandem Photovoltaics. <i>Science</i> , 2024, 383, eadh3849. DOI: 10.1126/science.adh3849
		6	Thiesbrummel, J., Shah, S., Gutierrez-Partida, E., Zu, F., Peña-Camargo, F., Zeiske, S., Diekmann, J., Ye, F., Peters, K. P., Brinkmann, K. O., Caprioglio, P., Dasgupta, A., Seo, S., Adeleye, F. A., Warby, J., Jeangros, Q., Lang, F., Zhang, S., Albrecht, S., Riedl, T., Armin, A., Neher, D., Koch, N., Wu, Y., Le Corre, V. M., Snaith, H., & Stolterfoht, M. Ion-Induced Field Screening as a Dominant Factor in Perovskite Solar Cell Operational Stability.

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		7	Park, J., Kim, J., Yun, H.-S., Paik, M. J., Noh, E., Mun, H. J., Kim, M. G., Shin, T. J., & Seok, S. I. Controlled Growth of Perovskite Layers with Volatile Alkylammonium Chlorides. Nature, 2023, 616, 724–730. DOI: 10.1038/s41586-023-05825-y
		8	Cao, F., Bian, L., & Li, L. Perovskite Solar Cells with High-Efficiency Exceeding 25%: A Review. Energy Materials and Devices, 2024, 2, 9370018. DOI: 10.26599/EMD.2024.9370018
		9	Luo, C., Zheng, G., Gao, F., Wang, X., Zhao, Y., Gao, X., & Zhao, Q. Recent Advances and Remaining Challenges in Perovskite Solar Cell Components for Innovative Photovoltaics. Nanomaterials, 2024, 14, 1867. DOI: 10.3390/nano14231867
		10	Nair, S., Sree Raj, R. K., Bharat, L. K., Joseph, B., & Kalarikkal, N. Perovskite Solar Cells: A Review of the Latest Advances in Materials, Fabrication Techniques, and Stability Enhancement Strategies. Micromachines, 2024, 15, 192. DOI: 10.3390/mi15020192
Claude (Sonnet 4.6)	QD LEDs	1	Jang, E.; Jang, H. Review: Quantum Dot Light-Emitting Diodes. Chemical Reviews, 2023, 123(8), 4663–4692. DOI: 10.1021/acs.chemrev.2c00695
		2	Li, B.; Chen, F.; Xu, H.; Song, Y.; Yan, X.; Xu, Q.; Wu, L.; Yan, Y.; Hou, W.; Cao, W.; Shen, H.; Fan, F. Advances in Understanding Quantum Dot Light-Emitting Diodes. Nature Reviews Electrical Engineering, 2024, 1, 412–425. DOI: 10.1038/s44287-024-00059-0
		3	Xu, H.; Song, J.; Zhou, P.; Song, Y.; Xu, J.; Shen, H.; Fang, S.; Gao, Y.; Zuo, Z.; Pina, J. M.; Voznyy, O.; Yang, C.; Hu, Y.; Li, J.; Du, J.; Sargent, E. H.; Fan, F. Dipole–Dipole-Interaction-Assisted Self-Assembly of Quantum Dots for Highly Efficient Light-Emitting Diodes. Nature Photonics, 2024, 18, 186–191. DOI: 10.1038/s41566-023-01344-4
		4	Fan, J.; Han, C.; Yang, G.; Song, B.; Xu, R.; Xiang, C.; Zhang, T.; Qian, L. Recent Progress of Quantum Dots Light-Emitting Diodes: Materials, Device Structures, and Display Applications. Advanced Materials, 2024, 36, 2312948. DOI: 10.1002/adma.202312948
		5	Chen, X.; Lin, X.; Zhou, L.; Sun, R.; Li, M.; Chen, Y.; Yang, W.; Hou, L.; Wu, W.; Cao, W. Blue Light-Emitting Diodes Based on Colloidal Quantum Dots with Reduced Surface–Bulk Coupling. Nature Communications, 2023, 14, 284. DOI: 10.1038/s41467-023-35853-9
		6	Zhang, W.; Li, B.; Chang, C.; Chen, F.; Zhang, Q.; Lin, Q.; Wang, L.; Yan, J.; Wang, F.; Chong, Y.; Du, Z.; Fan, F.; Shen, H. Stable and Efficient Pure Blue Quantum-Dot LEDs Enabled by Inserting an Anti-Oxidation Layer. Nature Communications, 2024, 15, 783. DOI: 10.1038/s41467-024-44894-z
		7	Bian, Y.; Yan, X.; Chen, F.; Li, Q.; Li, B.; Hou, W.; Lu, Z.; Wang, S.; Zhang, H.; Zhang, W.; Zhang, D.; Tang, A.; Fan, F.; Shen, H. Efficient Green InP-Based QD-LED by Controlling Electron Injection and Leakage. Nature, 2024, 635, 854–859. DOI: 10.1038/s41586-024-08136-w
		8	Du, W.; Cheng, C.; Tian, J. Efficient Solution-Processed InP Quantum-Dots Light-Emitting Diodes Enabled by Suppressing Hole Injection Loss. Nano Research, 2023, 16(5), 7511–7517. DOI: 10.1007/s12274-022-5268-4
		9	Gao, Y.; Huang, Y.; Liu, M.; Zhang, L.; Li, X.; He, Z.; Lin, Q. Minimizing Heat Generation in Quantum Dot Light-Emitting Diodes by Increasing Quasi-Fermi-Level Splitting. Nature

			Nanotechnology, 2023, 18, 1168–1174. DOI: 10.1038/s41565-023-01470-8
		10	Li, H.; Feng, Y.; Zhu, M.; Gao, Y.; Fan, C.; Cui, Q.; Cai, Q.; Yang, K.; He, H.; Dai, X.; Huang, J.; Ye, Z. Accurate Characterization of Quantum-Dot Light-Emitting Diodes for Display Applications. Nature Nanotechnology, 2024, 19, 638–646. DOI: 10.1038/s41565-024-01612-4
DeepSeek (DeepSeek-V3.2)	Electrochemical CO ₂ Reduction	1	Yuhang Jiang, et al. Electrochemical CO2 Reduction Using Membrane Electrode Assemblies: Progress, Challenges, and Opportunities. Chemistry – An Asian Journal, 2025, 20, e00685. DOI: 10.1002/asia.202500685
		2	Grazia Leonzio, Anna Hankin, Nilay Shah. CO2 electrochemical reduction: a state-of-the-art review with economic and environmental analyses. Chemical Engineering Research and Design, 2024, 208, 934-955. DOI: 10.1016/j.cherd.2024.07.014
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Responses Run 2 (March 19, 2026)			
Prompt: “Provide exactly 10 peer-reviewed journal citations on: new advances and challenges in (topic area)” Format each citation as: [Authors]. [Title]. [Journal], [Year], [Volume], [Pages]. DOI: [DOI].”			
ChatGPT (GPT 5.3)	Electrochemical CO ₂ Reduction	1	Ha Huu Do, Hai Bang Truong. Ni, Co, Zn, and Cu metal-organic framework-based nanomaterials for electrochemical reduction of CO ₂ : A review. <i>Beilstein Journal of Nanotechnology</i> , 2023, 14, 904–911. DOI: 10.3762/bjnano.14.74.
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Claude (Sonnet 4.6)	Electrochemical CO ₂ Reduction	1	Han, J.; Bai, X.; Xu, X.; Bai, X.; Husile, A.; Zhang, S.; Qi, L.; Guan, J. Advances and Challenges in the Electrochemical Reduction of Carbon Dioxide. <i>Chemical Science</i> , 2024, 15, 7870–7907. DOI: 10.1039/D4SC01931H
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Claude (Sonnet 4.6)	Li-ion Batteries	1	Frith, J.T., Lacey, M.J., Ulissi, U. A non-academic perspective on the future of lithium-based batteries. <i>Nature Communications</i> , 2023, 14, 420. DOI: 10.1038/s41467-023-35933-2
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Claude (Sonnet 4.6)	Perovskite Solar Cells	1	Liu, J., He, Y., Ding, L., et al. Perovskite/silicon tandem solar cells with bilayer interface passivation. <i>Nature</i> , 2024, 635, 596–603. DOI: 10.1038/s41586-024-07997-7
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Meta (Llama 4)	Li-ion Batteries	1	Makeyaw, A., Dame, T., Getu, M., Goytom, D., Tadese, M., et al. Targeted Advances in Lithium-ion Batteries: A Critical Review of Synergetic Improvements in Energy Density, Life Cycle, and Safety. <i>American Journal of Quantum Chemistry and Molecular Spectroscopy</i> , 2025, 9, 12-30. DOI: 10.11648/j.ajqcms.20250901.12
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		5	Rao, K. D., Rao, K. D., Pavani, P., Prasad, K. V. S. R., Indira, D., & Phaniteja, B. A critical review on lithium ion battery modeling, battery management system and thermal runaway issues. <i>Electrical Engineering</i> , 2025, NA, NA. DOI: 10.1007/s00202-025-03102-x
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		7	Zhang, Z., et al. Research progress of silicon-based anode materials for lithium-ion batteries. <i>RSC Adv</i> , 2025, 15, 10731-10753. DOI: 10.1039/d5ra01268f
		8	Parvizi, P., Jalilian, M., Amidi, A. M., Zangeneh, M. R., & Riba, J. R. From Present Innovations to Future Potential: The Promising Journey of Lithium-Ion Batteries. <i>Micromachines (Basel)</i> , 2025, 16, 194. DOI: 10.3390/mi16020194
		9	Li, J., Peng, Y., Wang, Q., & Liu, H. Status and Prospects of Research on Lithium-Ion Battery Parameter Identification. <i>Batteries</i> , 2024, 10, 194. DOI: 10.3390/batteries10060194
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			scales. Journal of Energy Storage, 2023, 64, 107145. DOI: 10.1016/j.est.2023.107145
Meta (Llama 4)	Perovskite Solar Cells	1	Mohammadi, M., Ji, F., Sachsenweger, T. et al. Integrated memristor for mitigating reverse-bias in perovskite solar cells. Nature, 2026. DOI: 10.1038/s41586-026-10275-3
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		3	Zheng, L., Zhu, H., Wei, J. et al. Strain-induced rubidium incorporation into wide-bandgap perovskites reduces photovoltage loss. Science, 2025, 388, 88–95.
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		9	Yu, S., Li, Z., Sun, Z. et al. Homogenized NiO x nanoparticles for improved hole transport in inverted perovskite solar cells. Science, 2023, 382, 1399–1404.
		10	Zhang, D., Li, D., Hu, Y., Mei, A. & Han, H. Degradation pathways in perovskite solar cells and how to meet international standards. Communications Materials, 2022, 3, 58.
Meta (Llama 4)	QD LEDs	1	Jaehoon Kim. Recent progresses and challenges in colloidal quantum dot light-emitting diodes: a focus on electron transport layers with metal oxide nanoparticles and organic semiconductors. Nanoscale Horizons, 2024, 9, 2167-2197. DOI: 10.1039/D4NH00370E
		2	Aqiang Liu, Chunyan Cheng, Jianjun Tian. Exploring performance degradation of quantum-dot light-emitting diodes. Journal of Materials Chemistry C, 2022, 10, 8642-8649. DOI: 10.1039/D2TC01355J
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		9	Recent Advances and Challenges of Colloidal Quantum Dot Light-Emitting Diodes for Display Applications. Advanced Materials, 2024, 36, e2212220. DOI: 10.1002/adma.202212220
		10	Blue Quantum Dot Light-Emitting Diodes toward Full-Color Displays: Materials, Devices, and Large-Scale Fabrication. Nano Letters, 2025, 25, 1-15. DOI: 10.1021/acs.nanolett.4c02968

Table S2

Repetition of same citations across different LLMs

Metric	Run 1	Run 2	Notes
Total citations loaded	200	200	All rows in each run file
Multi-engine duplicates (same day)			
No. of Articles -DOIs cited by >1 engine	15	18	Same DOI, different engines, same run day
Total citation count	35	39	Total rows in multi-engine groups
Total fraction of duplicates	18.5%		Sum of total duplicate citations divided by total (400)

Multi-engine duplicates highlight repetition with respect to number of citations and total number of citations across five different LLMs.

Flowchart S1

Process of citation validation using CrossRef API. Results were also checked manually prior to running additional analyses

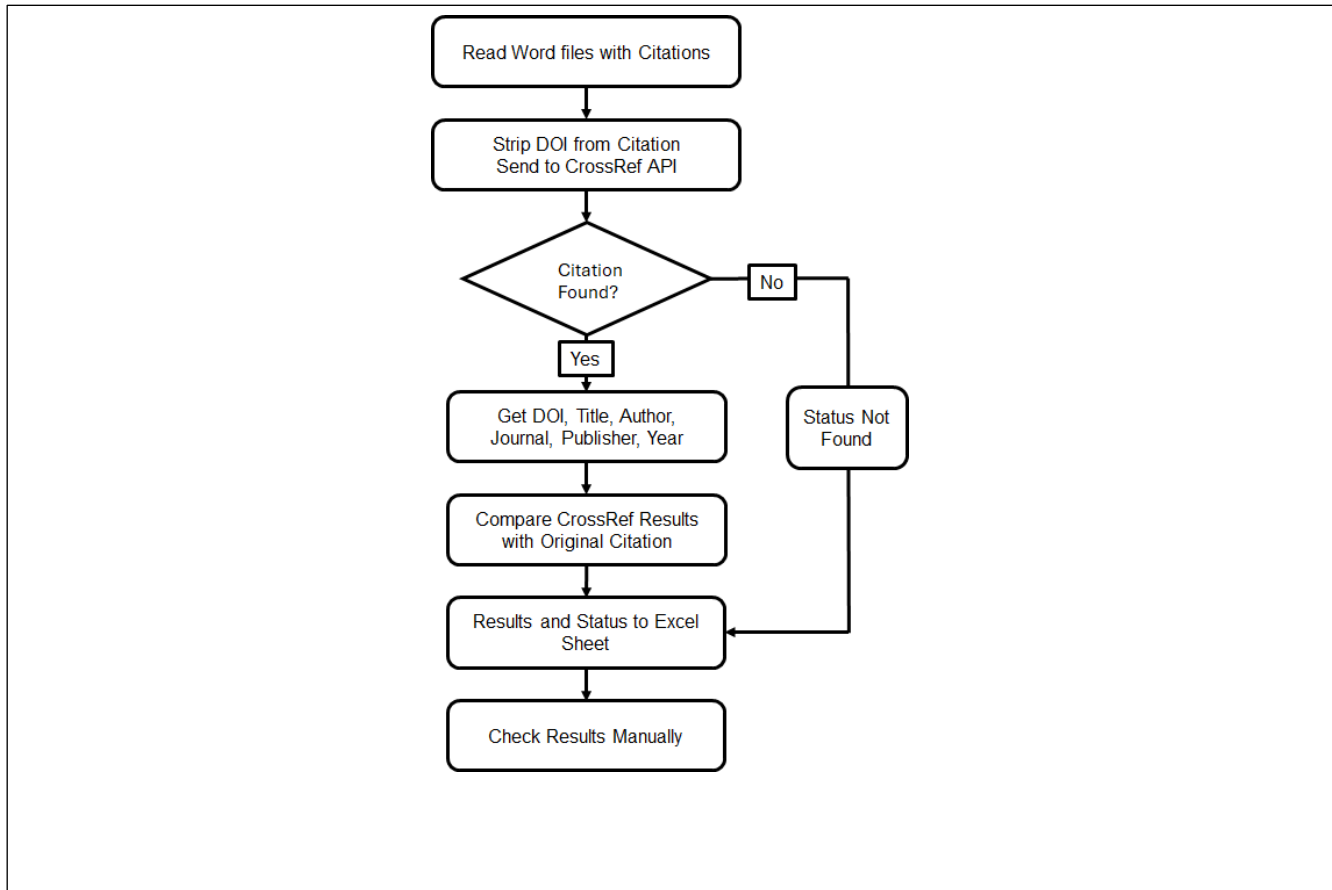


Table S3

Comparison of invalid citations generated by LLMs during two runs

AI Engine	Total	Invalid Run #1	Invalid % Run #1	Invalid Run #2	Invalid% Run #2	Invalid Cites Total	Invalid Cites (Average) %
ChatGPT	40+40	9	22.5%	3	7.5%	12	15.0%
Claude	40+40	1	2.5%	1	2.5%	2	2.5%
DeepSeek	40+40	6	15.0%	8	20.0%	14	17.5%
Gemini	40+40	4	10.0%	4	10.0%	8	10.0%
Meta	40+40	2	5.0%	0	0.0%	2	2.5%
TOTAL	200+200	22	11.0%	16	8.0%	38	9.5%

Table S4

Top five journals cited by each LLM (based on only valid citations)

	ChatGPT	Claude	DeepSeek	Gemini	Meta
Journal (Count)	RSC Advances (5)	Nature (9)	Nature Communications (5)	Nanomaterials (5)	Advanced Materials (4)
	Nature Communications (5)	Nature Communications (7)	ACS Applied Energy Materials (5)	Advanced Materials (4)	Nature Communications (4)
	Nano-Micro Letters (3)	Nature Energy (5)	Advanced Materials (4)	Energy Materials (3)	Journal of the American Chemical Society (3)
	Micromachines (3)	Science (5)	Chemistry – An Asian Journal (2)	Nano-Micro Letters (3)	RSC Advances (3)
	Advanced Materials (3)	Frontiers in Batteries and Electrochemistry (4)	Chemical Engineering Research and Design (2)	Batteries (3)	Micromachines (3)